

Training Course

SonarQube



What and Why SonarQube?

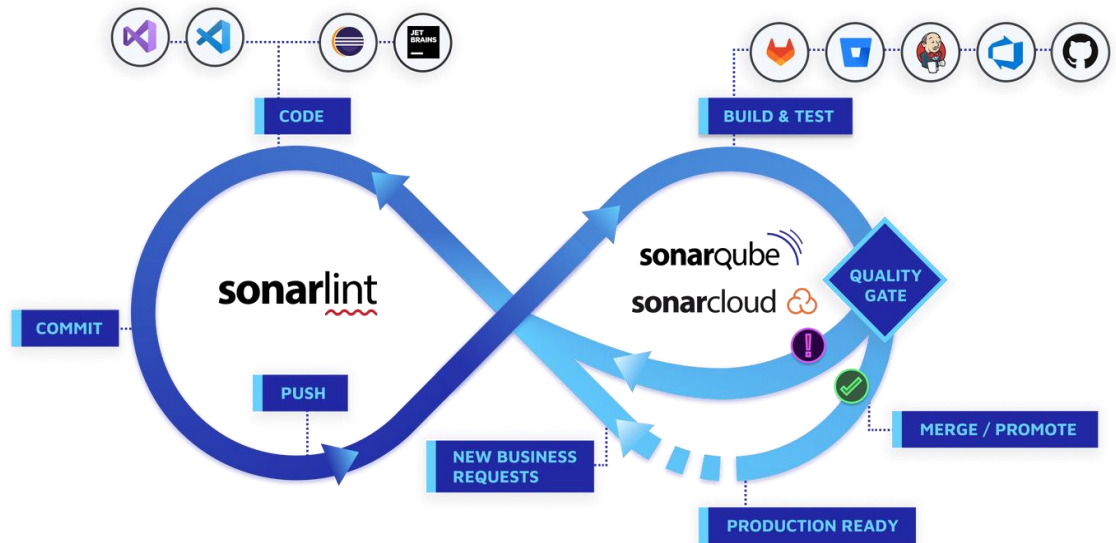


- Self-managed, automatic code review tool that helps deliver clean code.
- Helps in identifying and fixing code vulnerabilities, bugs.
- Supports 30+ programming languages: source code, infra as code, test code, scripts, ...

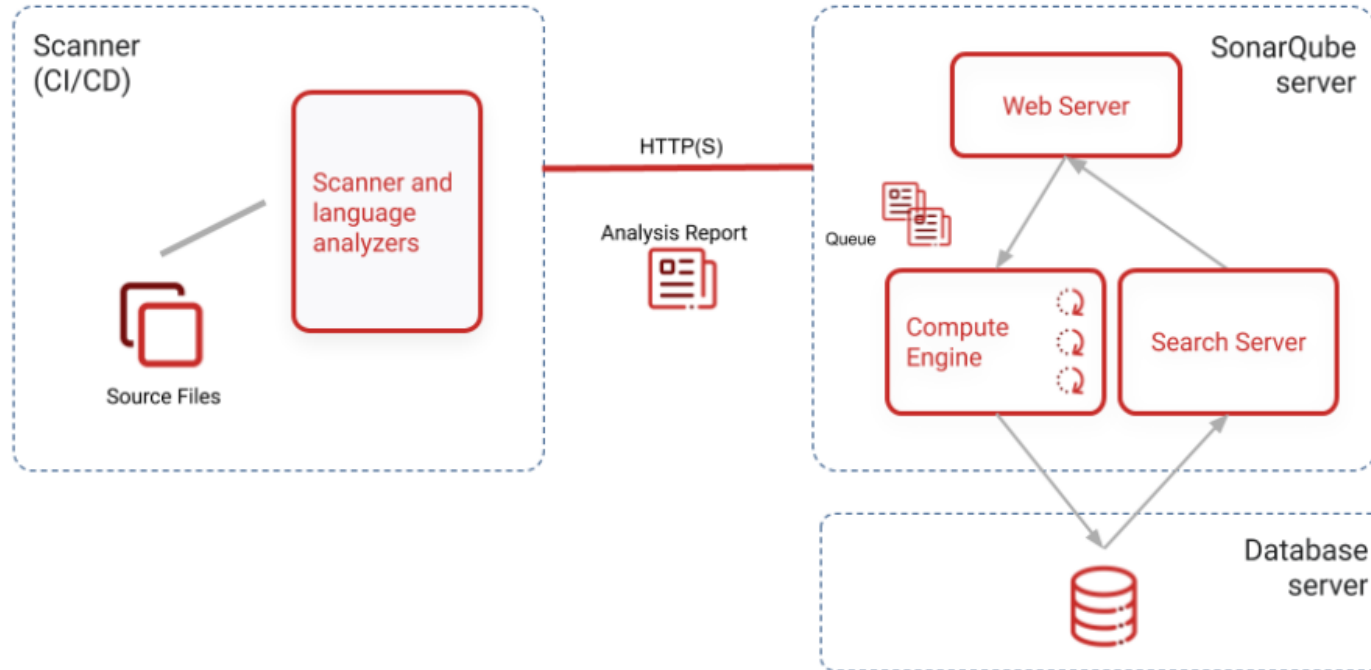


Sonar solutions

- SonarLint: provides immediate feedback in IDE.
- SonarQube: integrates in CI pipeline to detect issues.

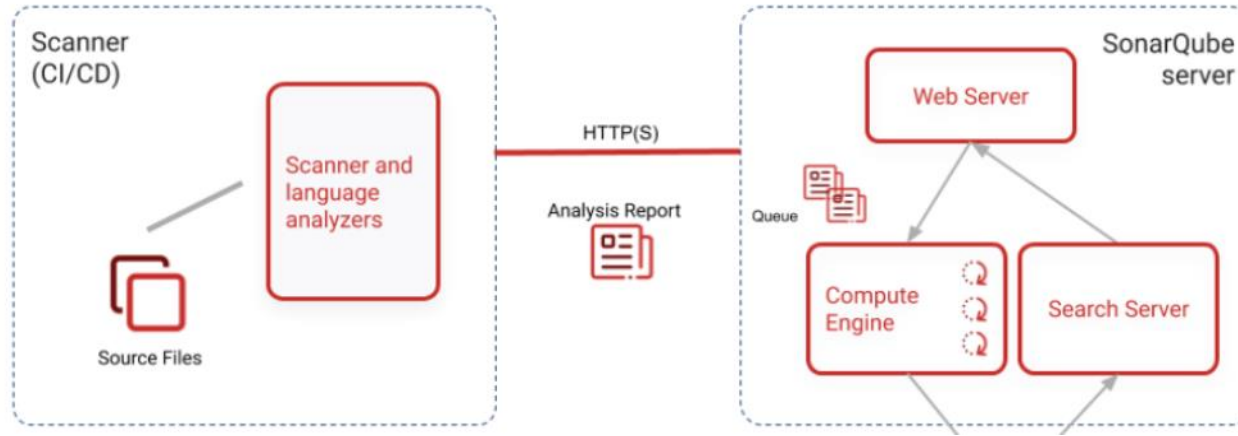


SonarQube components



SonarQube scanner

- Scan source files.
- Send analysis report to SonarQube server.



SonarQube scanner

- Configured by file sonar-project.properties.

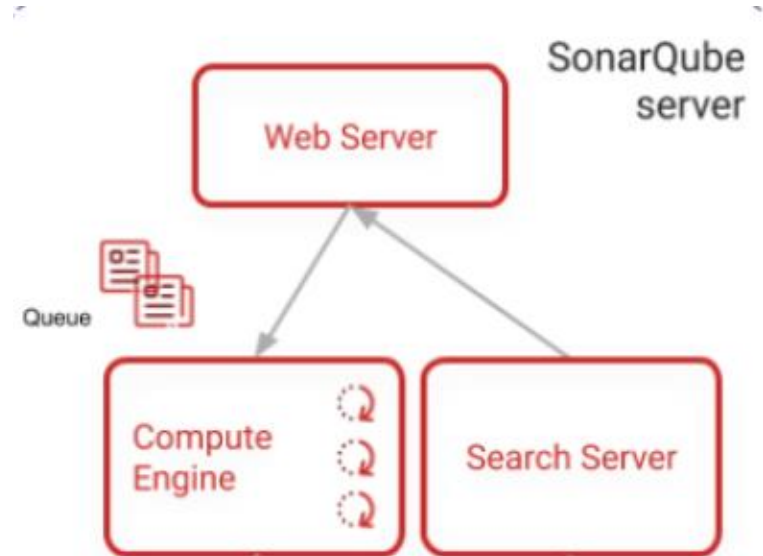
```
sonar.projectKey = <unique key>  
sonar.sources = <relative location to the source code>
```

- Or, by specifying properties using command line.

```
Sonar-scanner -Dsonar.projectKey=<unique key>
```

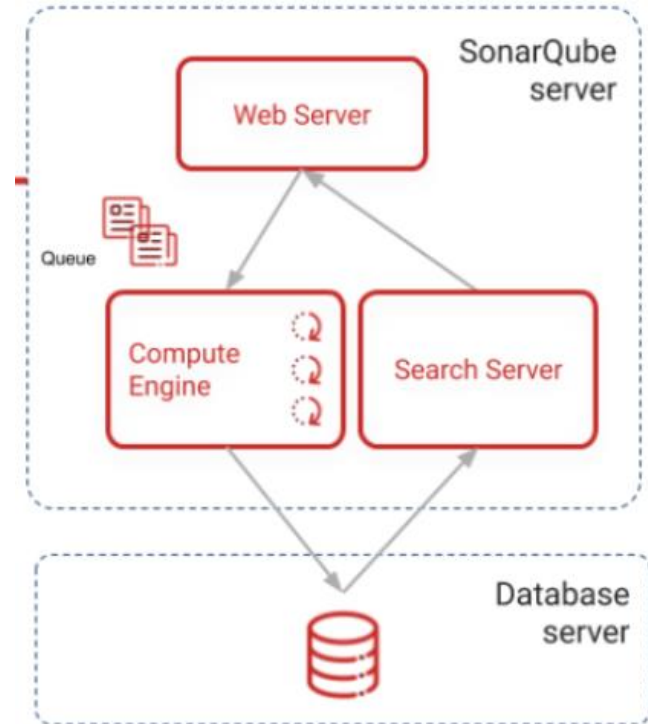
SonarQube server

- A web server that user interface.
- A search server based on ElasticSearch.
- Compute engine processes the code analysis and saves to database.



SonarQube database server

- Metrics and issues for code quality and security generated during code scans.
- The SonarQube instance configuration.



- Integrates with popular DevOps tools and workflows, such as Jenkins, GitLab, and Azure DevOps.



- It can be integrated into CI/CD pipelines to perform code analysis automatically at each build stage.
- Plugins and extensions are available for various IDEs.



Thank you

